LAB EXCEPTION ACTIVITY ANSWERS

ACTIVITY A.

QUESTION 1

Integer.parseline(“ABC”)

- It throws a NumberFormatException because "ABC" contains alphabetic characters rather than a valid integer.
    - Exception: NumberFormatException
    - Classification: Unchecked exception.
    - Explanation: Java expects a valid numeric string, but "ABC" contains letters, so it cannot be converted into an integer.
    - Prevention/handling strategy: Validate the input before conversion or use a try-catch block to catch NumberFormatException.

QUESTION 2

String name = null; name.length();

- This code throws a NullPointerException because the variable name contains null, meaning it does not refer to any String object.
- The program cannot call the length() method on a null reference.
    - Exception: NullPointerException
    - Classification: Unchecked exception.
    - Explanation: The variable name does not refer to an actual String object. Calling length() on null causes an exception.
    - Prevention/handling strategy: Check whether the variable is null before calling its methods.

QUESTION 3

new java.io.FileReader("missing.txt");

- This code throws a **F**ileNotFoundException if the file missing.txt does not exist or cannot be opened.
- Java cannot create a FileReader for a file it cannot access.
    - Exception: FileNotFoundException
    - Classification: Checked exception.
    - Explanation: Java cannot find or open the specified file.

- Prevention/handling strategy: Check whether the file exists before opening it, or handle the exception using try-catch or declare it using throws.

QUESTION 4

- OutOfMemoryError occurs when the Java Virtual Machine cannot allocate the memory required by the application. Catching it and simply continuing may cause further failures because the application may still lack sufficient memory to operate safely.

6. ACTIVITY D

QUESTION 1.

- `IOException` is a checked exception. The `throws` keyword appears in `importMarks()` because file-reading operations may cause input/output errors that must be handled by the calling method.

QUESTION2.

- throw is used to raise an exception, for example: throw new DuplicateStudentException("Duplicate student ID: " + studentID).
- throws declares exceptions a method may pass to its caller, for example: static void importMarks(String filename) throws IOException.

QUESTION3.

- Friendly messages help users understand errors, while the logger stores technical details to help programmers investigate and fix problems.

QUESTION4.

- An empty catch block silently ignores errors, making debugging difficult. Catching `Exception` before `FileNotFoundException` causes a compilation error because the general exception catches the specific one first.